Haoran Yin

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Education

University of Toronto — Toronto, Canada
Ph.D in Aerospace Science & Engineering — *Sep 2026 – Nov 2030*

Georgia Institute of Technology — Atlanta, USA
M.Sc. in Robotics; GPA: 4.00/4.00 — *Aug 2024 – May 2026*

University of Toronto — Toronto, Canada
B.A.Sc. in Computer Engineering; GPA: 3.88/4.00 — *Sep 2019 – Jun 2024*
Minor in Robotics and Mechatronics — *Graduated with High Honor (Top 10%)*

Selected Publications

*: Equal contribution; ranked in alphabetical order.

[1] Devesh Nath*, Anutam Srinivasan*, **Haoran Yin***, Ruitong Jiang, Jeffrey Fang, Glen Chou, "Pixels to Proofs: Probabilistically-Safe Latent World Model Control via Parallel Conformal Robust MPC," *RA-L, in submission*, 2026.

[2] Devesh Nath*, **Haoran Yin***, Glen Chou, "Scalable Data-Driven Reachability Analysis and Control via Koopman Operators with Conformal Coverage Guarantees," *L4DC*, 2026.

[3] Devesh Nath*, **Haoran Yin***, Glen Chou, "Formal Safety Verification and Refinement for Generative Motion Planners via Certified Local Stabilization," *ICRA*, 2026.

Awards & Achievements

Best Paper Award Finalist, L4DC 2026 Top 3 research paper out of 106 accepted submissions

Research Projects

Toronto Robotics and AI Lab — Toronto, Canada
Doctoral Researcher, Supervisor: Dr. Steven Waslander — *Sep 2026 – Present*

Teaching Robots to Learn from Their Mistakes: Agentic Recovery and Continual Fine-Tuning of VLA Policies (On-going)

- Developing an agentic framework in which a VLM coding agent decomposes long-horizon tasks for an open-weight VLA policy (e.g., $\pi_{0.5}$) and, when a failure is detected, writes executable recovery programs using perception and motion-planning primitives.
- Designing a continual fine-tuning pipeline that distills agent-recovered episodes into the VLA's weights, evaluated on a continual-learning benchmark that measures intervention decay and skill retention.

Steering Robots Before They Fail: Predictive Activation Steering for Safe VLA Policies (On-going)

- Developing lightweight probes and a latent dynamics model that predict how failure risk and physical quantities decoded from a frozen VLA's internal activations evolve under candidate steering inputs.
- Designing a predictive steering controller that plans the smallest activation edit over a short horizon to keep predicted risk below a conformally calibrated threshold, preventing failures without retraining.

Trustworthy Robotics Lab — Atlanta, USA
Graduate Researcher, Supervisor: Dr. Glen Chou — *Dec 2024 – Aug 2026*

Pixels to Proofs: Probabilistically-Safe Latent World Model Control via Parallel Conformal Robust MPC

- Led the experimental implementation for Pixels to Proofs outside of the SLS-based robust MPC formulation.
- Trained latent world models and implemented nominal iLQR / gradient-based MPC planning across Reacher, Cube, Push-T, and bimanual rope manipulation.
- Implemented and evaluated baseline comparisons against prior latent world-model and safe-planning methods, including Le-WM, DINO-WM, PLDM, Hamilton-Jacobi-filtered planning, and Latent Policy Barrier.
- Built the real-world bimanual rope manipulation hardware setup and conducted hardware experiments for visual rope manipulation.

Scalable Data-Driven Reachability Analysis and Control via Koopman Operators with Conformal Coverage Guarantees

- Trained Deep Koopman models for multiple dynamic systems, designing custom loss functions to achieve near-perfect dynamics reconstruction.
- Designed a time-dependent LQR controller in the latent space to enable end-to-end comparison between latent and original-space rollouts.
- Co-designed a framework integrating conformal prediction and formal verification, achieving 100% empirical coverage while reducing computational time and conservativeness by orders of magnitude.
- Collected datasets and validated performance by conducting experiments on diverse systems, including multi-DoF quadrotors and high-dimensional MuJoCo environments (up to 28D).

Formal Safety Verification and Refinement for Generative Motion Planners via Certified Local Stabilization

- Authored the original CasADi-based data collection pipeline for all analytic/learned systems (e.g., unicycle, car, 2D/3D quadrotor), and implemented the MuJoCo dataset script for the rope task.
- Co-designed the full framework—training routines, algorithms, and verification workflow—used in the paper.
- Rapidly implemented and benchmarked strong baselines (incl. Neural CBF) with clean, reproducible training/eval scripts.
- Improved and debugged the formal verification stack (AutoLiRPA/CROWN reachability) via extensive code reviews and fixes.

Working Experience

Georgia Institute of Technology Atlanta, USA

Graduate Teaching Assistant — CS 6601: Artificial Intelligence *Aug 2025 – Dec 2025*

- Monitor and promptly reply to student questions on the course forum, clarifying concept.
- Design exam questions, author official solutions and rubrics, and assist with proctoring and grading.

Intel Corporation San Jose, USA

Software Engineering *May 2022 – April 2023, Internship*

- Spearheaded the independent development of firmware for two FPGA I/O units, coupled with rigorous analysis of port simulation data to validate performance and ensure design integrity.
- Performed detailed analysis of Hardware Memory Controller (HMC) simulations to evaluate system performance and verify adherence to design specifications.
- Designed and implemented a Perl-based automation framework for HMC simulations, enhancing testing efficiency and workflow scalability.

Conceptual Strengths and Technical Skills

Programming Skills: C, C++, Python, MATLAB, ROS/ROS2, Git, Perforce, Assembly Language

Robotics Modelling and Control: Convex Optimization, Lagrangian Modelling, Linear and Nonlinear Control, Model Predictive Control

Machine Learning: Generative Model (Flow Matching, Diffusion)

Deep Reinforcement Learning: Policy Gradient, Actor-Critic, Imitation Learning, Multi-Agent Deep Deterministic Policy Gradient, Model-based RL

Computer Vision: OpenCV, Image Processing, Object Detection, Scene Representation (NeRF, Guassian Splatting), Semantic Segmentation